

Alexander Totah

alex.totah278@gmail.com | +1(714) 897-4665 | linkedin.com/in/alexander-totah | nander100.github.io

About Me

AFRL Sensors Directorate research intern and UC Riverside EE graduate building real-time autonomous systems seeking full-time engineering roles in defense. Track record of hitting hard performance targets – sub-30ms latency, 95%+ accuracy – and leading technical teams.

Technical Experience

Research Intern, Air Force Research Lab (Sensors Directorate Internship Program) May 2026 – Present

- Artificial intelligence/neural networks, Modeling and simulation (*Technical details pending public release*)
- Expected publication Oct. 2026

Autonomous Vehicle Engineer, Collaborative Intelligence Systems Lab Jan. 2024 – Jun. 2026

- Coordinated development of an imitation learning pipeline using **ROS** and **Docker** with **TensorFlow** path planning algorithms, reducing lap times by 1.6 seconds over baseline.
- Optimized sensor data processing by using **Python** (PyTorch) autoencoder that compressed LiDAR data by 94% while maintaining 96% accuracy, enabling real-time operation
- Developed **PID-controlled** brushless motor system, achieving simulation-validated performance.

Systems and Control Engineer, Robotics and Medical Systems Lab Oct. 2022 – Jan. 2024

- Designed a **PID control** system using **MATLAB** to control a steerable needle with sub-1 ms latency.
- Engineered **Arduino-based** controls for an MRI-compatible biopsy robot, reducing user error during procedures.
- Calibrated fiber optic sensor using **Simulink** to measure the bend of a steerable needle within less than 0.5mm.

Projects

Simple Arm: Camera-Controlled Robotic Arm Project Repo

- Designed and built a 4-DOF robotic arm: developed joint control, sensor feedback, and gripper actuation.
- Optimized network communication pipeline, cutting latency from 35 ms to 26 ms to meet a critical requirement.
- Conducted a usability study; measured a 25 s mean time-to-grasp, validating task efficiency for first-time users.

Plock - Signature Based Penlock Project Repo

- Achieved 95% accuracy on an SVM-based signature lock system with zero forged signatures accepted.
- Engineered system model and developed backend SVM signature validation using **Python**; earned 3rd place overall out of 42 teams.

Leadership Experience

Co-Founder and Club President, Combat Robotics Club May 2023 – May 2025

- Secured \$5,000+ in sponsorship funding from university donors, enabling expansion of robotics initiatives.
- Organized an annual competition showcasing student-created robots with 40+ spectators.

Skills

Software & Programming: C++, Python, MATLAB, Simulink, Ubuntu/Linux CLI, Arduino, SolidWorks, Docker

Circuits & Embedded Systems: Arduino, AVR Microcontrollers, Raspberry Pi, Altium, PSpice, I2C, SPI, USART

AI & Robotics: PyTorch, TensorFlow, ROS/ROS2, Nvidia Jetson Platform, Keras, OpenAI API

Education

UC Riverside, **B.S. Electrical Engineering, Computer Science Minor**, 3.5/4.0

Jun. 2026